Amendment 1 — engPENK peptide identity and granule processing

Supersedes: §1 "Three ordering details that are easy to get wrong" of `ENGPENK_PEPTIDE_WETLAB.md` (2026-08-06). Everything else in that sheet stands. **Construct:** `engPENK_therapy_plasmid`, 7× `YGGFEVVG` cassette in pcDNA3.1(+). **Date:** 2026-08-16. **Basis:** sequence analysis of `engPENK_insert_annotated.gb` vs native PENK (P01210). **Status:** dry-lab finding. Nothing here has been measured in-house.

0. Why you are getting this sheet

The original reference sheet told you to order the **free acid** because the proprotein releases the peptide by proteolysis, which leaves a free carboxylate. That reasoning is correct for the convertase step and it is **incomplete**: the cassette ends in **Gly**, and the same granules that do the cutting also contain **PAM** (peptidylglycine α-amidating monooxygenase). PAM consumes a C-terminal Gly and amidates the residue in front of it.

Native PENK cores end in Met or Leu (`YGGFM`, `YGGFL`) and are therefore **not** PAM substrates. This is the one respect in which the engineered cassette behaves differently from native PENK, and it changes the identity of the product.

```
cassette as built:   ...KR-YGGFEVVG-KR...
  convertase + CPE ->  YGGFEVVG        free acid, 827.3934 [M+H]+
  then PAM         ->  YGGFEVV-NH2     769.3879 [M+H]+   <-- 7-mer, amide on VAL
```

The MD and docking work in this project scored a **deltorphin-II-like** ligand, which is `Y-a-F-E-V-V-G-NH2` — amide on **Gly**. The cassette as built cannot produce that geometry.

1. Decision required before any cloning proceeds

Option	Cassette	Granule product	Consequence
A — leave as built	`YGGFEVVG`	`YGGFEVV-NH₂` (769.3879)	7-mer, amide on Val. Not the species that was modelled; the δ-selectivity predictions do not transfer without re-scoring.
B — add one Gly	`YGGFEVVGG`	`YGGFEVVG-NH₂` (826.4094)	Matches deltorphin-II C-terminal geometry. **One codon.** Cheap now, expensive after cloning.

This is a design decision, not a bench decision — but it gates the peptide order, so it has to be made first. If B is chosen, the insert must be re-synthesised and the QC protocol fragment sizes in `engPENK_therapy_qc_protocol.md` shift by 3 bp per cassette (21 bp total).

2. Corrected peptide order

Order **three** peptides, not two. All all-L, no D-residues anywhere.

#	Sequence	C-terminus	Monoisotopic $[M+H]^+$	Why you need it
1	`YGGFEVVG`	**free acid**	827.3934	unamidated fraction — the species already docked/scored
2	`YGGFEVV`	**amide**	769.3879	the PAM product from the cassette as built
3	`YGGFM`	free acid	574.2330	native Met-enkephalin comparator, and present in your cells

If option B is taken, add `YGGFEVVG`-**NH_2** (826.4094) as #4.

> **Vendor trap — this is where the 0.98 Da bites.** Free acid and amide of the same sequence differ by only 0.98 Da. The amide's first ^{13}C peak (827.4127) sits 0.0193 Da from the free acid's monoisotopic peak (827.3934) — **R ≈ 42,784 FWHM** to separate them. Many vendors default to C-terminal amide. Specify the terminus in writing on the order, and QC the delivered standard on a high-resolution instrument (Orbitrap or high-end TOF) — a quadrupole or ion trap **cannot** tell you which one arrived.

3. What to expect in transfected-cell supernatant

Five species from the construct as built. Within a single construct the amidation readout is a clean **58.01 Da** shift — unambiguous on any instrument. The high-resolution requirement above applies to standards, not to this measurement.

Species	$[M+H]^+$	Origin
`YGGFEVV-NH₂`	769.3879	cassettes 1,2,3,5,6 after PAM
`YGGFEVVG`	827.3934	cassettes 1,2,3,5,6, PAM-incomplete fraction
`YGGFEVVGRF`	1130.5629	**cassette 7** — monobasic flank, extended
`YGGFEVVGRGL`	1153.6000	**cassette 4** — monobasic flank, extended
`YGGFM`	574.2330	endogenous native Met-enkephalin

Cassettes 4 and 7 will not give clean product. All seven cassettes have a dibasic KR upstream, but 4 and 7 sit at the native positions of Met-enkephalin-Arg-Gly-Leu and -Arg-Phe and inherit those monobasic downstream flanks. CPE removes the Arg but not Gly-Leu or Phe, so roughly **2 of 7** of the yield is C-terminally extended and pharmacologically distinct. Expect these peaks; they are not contamination. Fixing them means giving cassettes 4 and 7 their own dibasic downstream flank.

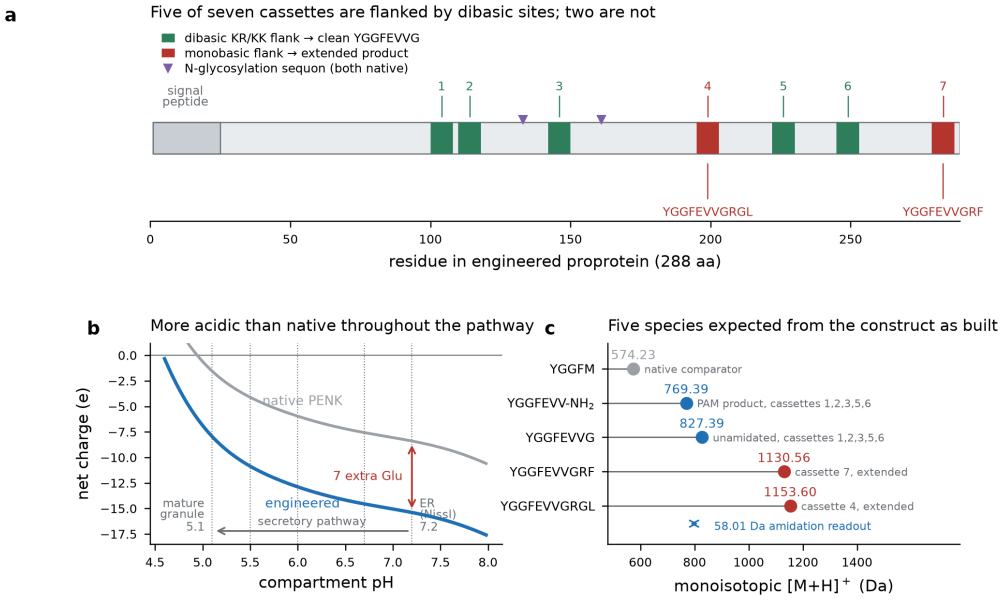


Figure 1. (a) Cassette map of the 288-aa engineered proprotein: five cassettes are flanked by dibasic KR/KK sites and give clean `YGGFEVVG`; cassettes 4 and 7 carry monobasic downstream flanks inherited from the native extended-peptide positions and give extended products. Both N-glycosylation sequons are native, displaced only by the upstream insertions. (b) Net charge vs. compartment pH along the secretory pathway — the engineered chain holds ~7 e more negative than native throughout (prediction, see §5). (c) The five peptide species expected from the construct as built, with the 58.01 Da amidation readout. All values computed from the annotated insert sequence.

4. What does not need changing

Verified against native PENK, computed from the sequences:

- **Signal peptide** residues 1–24, removed co-translationally. Mature chain 264 aa.
- **Disulfides:** 6 Cys in the mature chain → **3 disulfides, same count as native**. The cassettes contain no Cys, so they cannot scramble native pairing.
- **N-glycosylation:** two sequons, N133 (NGS) and N161 (NSS) — both **native** sequons, displaced only by the upstream insertions. No new glycosylation liability.
- **Where cleavage happens:** trans-Golgi onward, not the ER. PC1/3 and PC2 (with 7B2) are themselves pro-enzymes in the ER and only mature in the acidic, high-Ca^{2+} TGN. The proprotein transits the rough ER intact — which is what a 7-cassette tandem construct needs.

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5. One thing worth measuring that nobody has asked for yet

Each cassette replaces a Met-containing native core with an acidic one, so the mature chain carries **41 Glu against 34** in native: **−15.4 e vs −8.4 e at pH 7.2**, holding ~7 e more negative at every compartment pH along the pathway.

Dense-core granule sorting runs on pH- and Ca^{2+}-dependent self-aggregation in the TGN. Extra carboxylates could bridge Ca^{2+} better or raise solubility and resist condensation — it could go either way. **This is a prediction, not a result.** The experiment is cheap:

1. Transfect engPENK and native-PENK in the same cell line, same passage.
2. Split supernatant vs. cell lysate; quantify peptide in both by the MS method above.
3. Compare the **secreted : cell-associated** ratio between constructs.
4. Check for constitutive (unregulated) release — secretion without a secretagogue.

A construct that mis-sorts will show up as high constitutive release and low cell-associated peptide, and that would be a formulation problem, not a peptide problem.

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6. Bench checklist

- ☐ **Design decision A or B made and recorded** (§1) — blocks everything else
- ☐ Peptide order placed with **C-terminus specified in writing** (§2)
- ☐ Delivered standards QC'd on high-resolution MS before use (§2)
- ☐ MS method includes all five expected species, not just the octapeptide (§3)
- ☐ Sorting experiment scheduled with native-PENK control in the same cell line (§5)
- ☐ If option B: insert re-synthesised, QC fragment sizes recomputed (§1)